

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 19)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Two insecticides, X and Y, were tested on a pest insect and on a predatory ground beetle that feeds on the pest. Table 1 shows the results.

Table 1

Insecticide	LD ₅₀ for pest / mg kg ⁻¹	LD ₅₀ for ground beetle / mg kg ⁻¹	Half-life in soil / days	Solubility in water
X	5	6	180	Low
Y	8	150	4	High

0 1

. 1

Explain what is meant by LD₅₀.

[1 mark]

0 1

. 2

Using Table 1, explain which insecticide is more likely to cause indirect effects on the pest population through the food chain.

[3 marks]

0 1

. 3

Suggest **two** environmental problems that could be caused by insecticide Y that are less likely with insecticide X.

[2 marks]

1

2

0 1

. 4

Explain why the half-life of insecticide X makes it more likely to bioaccumulate.

[2 marks]

0 1 . 5

Give **two** reasons why a systemic insecticide may be preferred to a contact insecticide.

[2 marks]

1

2

10

0 2

A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 2 . 1

Use the formula $D = N(N - 1) / \sum n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

0 2 . 2

Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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.....

5

0 3

Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1

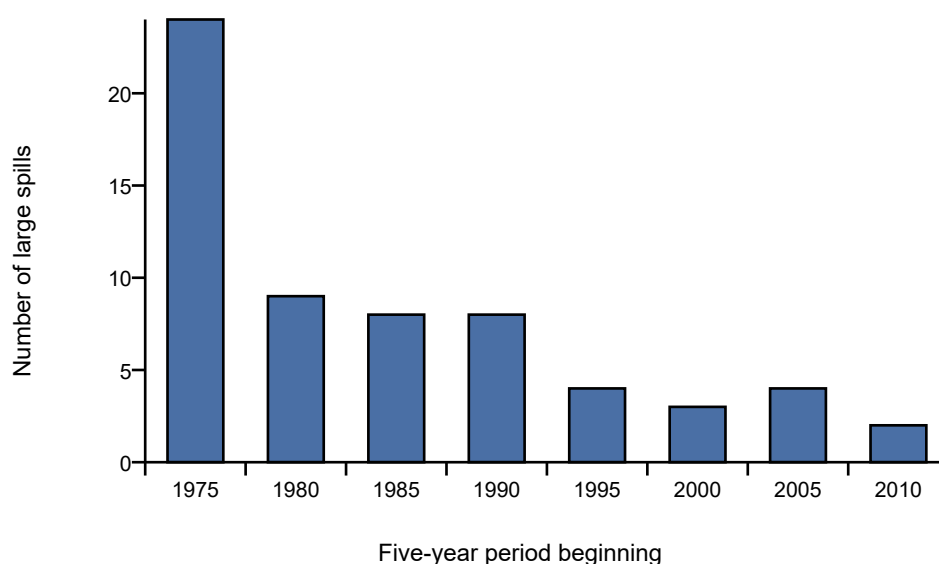


Table 1

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

0 3 . 1

Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]

Answer

%

03 . 2

Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

.....

.....

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.....

03 . 3

Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

1

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2

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3

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03 . 4

Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

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10

0 4 Table 1 shows the water quality required for four different uses.

0 4 . **1** Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason	
Potable (public) supply			
Spray irrigation			
Power station condenser cooling			
Industrial boiler water			

0 4 . **2** Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

0 4 . **3** Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

0 4 . 4

Give **two** methods used to sterilise treated water.

[2 marks]

1

2

10

0 5 . 1

Explain how integrated manufacture saves energy in the iron and steel industry.

[2 marks]

0 5 . 2

Explain why recycling is not always energy efficient.

[2 marks]

0 5 . 3

Explain how each of the following manages electricity infrastructure to conserve energy.

[4 marks]

High-voltage grid

ICT management of supply and demand

0	5	.	4
---	---	---	---

Give **two** locational factors affecting where new generating infrastructure is built.

[2 marks]

10

0	6	.	1
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Explain how climate change may affect the survival of species. Use named examples.

[6 marks]

0 6 . 2

Explain why a species living in an interdependent community may be unable to move to a new area even if suitable conditions exist there.

[2 marks]

0 6 . 3

Explain why polar regions warming faster than the equator affects the weather in the UK.

[2 marks]

10

0 7 . 1

Give **three** sources of mercury pollution.

[3 marks]

1

2

3

0 7 . 2

Describe **three** methods used to control mercury pollution.

[3 marks]

1

2

3

0 7 . 3 Explain why heavy metal wastes are stored under alkaline conditions. **[2 marks]**

0 7 . 4 State what is meant by synergism and give an example involving a heavy metal. **[2 marks]**

10

0 8 Table 1 shows the per capita energy consumption and GDP per capita of five countries.

Table 1

Country	GDP per capita / US\$ thousand	Energy use per capita / GJ per year	Main energy resource
A	62	290	Oil, gas
B	44	125	Gas, nuclear, renewables
C	10	95	Coal
D	2	25	Biomass (wood), oil
E	55	220	Hydroelectricity, oil

08 . 1

Describe the relationship between GDP per capita and energy use per capita shown in Table 1.

[2 marks]

08 . 2

Explain why affluence increases per capita energy consumption.

[3 marks]

08 . 3

Suggest **two** reasons why country B uses much less energy per capita than country A despite a similar level of affluence.

[2 marks]

1

2

08 . 4

Explain why the per capita energy consumption of the UK has fallen in recent years even though consumption of goods has risen.

[2 marks]

08 . 5

Give **one** way energy is used in each of the following: water supplies; agriculture.

[1 mark]

10

09 . 1

Describe how an **electrostatic precipitator** removes smoke particles from the effluent gases of a coal-fired power station.

[3 marks]

09 . 2

Explain how a **cyclone separator** works.

[2 marks]

09 . 3

Explain why heating coal to remove tar reduces smoke emissions.

[1 mark]

09 . 4

Explain why turbochargers reduce the smoke produced by diesel engines.

[2 marks]

0	9	.	5
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Explain why smaller smoke particles (PM₁) are a greater health risk than larger ones.

[2 marks]

10

1	0	.	1
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Evaluate the risks and benefits of the uses of ionising radiation, and explain the principles used to control exposure of workers and the public.

[9 marks]

Do not write outside the box

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Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

Do not write outside the box

END OF QUESTIONS